Jupiter

Software Valuation & Technical Due-Diligence Report

Software Valuation & Technical Due-Diligence Report

Date:	2026-06-30
Repository:	HenrikFaul/Jupiter
Version:	v0.1.0
Language:	English
Confidence:	Medium

Table of Contents

1. Executive Summary

Jupiter is a **native, privacy-first Android file manager** (single *:app* module, *com.jupiter.filemanager*) that aims to be **as simple as Files by Google but as powerful as Solid Explorer / X-plore**. It pairs everyday local file management with **real LAN/remote backends** (SMB, SFTP, FTP, WebDAV), an embedded **Wi-Fi desktop-transfer server**, an **AES-GCM encrypted vault** with biometric unlock, broad **archive support** (zip/tar/gz/7z/rar), **WorkManager automation**, an in-app PDF/media viewer, and an **optional on-device Claude (Anthropic) AI assistant** gated behind a user-supplied key. The product is explicitly **no-ads, no-dark-patterns**. It targets power users, creators, office/students, privacy-conscious users, and LAN/NAS users. Stage: **feature-complete pre-revenue MVP** — no users yet, not monetised.

Key Statistics (from repository scan)

212	33,932	237	~27	12	16
Kotlin Files	Lines of Kotlin	Total Files	Feature Packages	Domain Repositories	Git Commits

Build Effort Summary

Metric	Low	Most Likely	High
Person-hours	2,106	3,159	4,940
Person-months (160 h/mo)	13.2	19.7	30.9
Calendar months (2–3 person team)	5	7	11

Build Cost Summary

Scenario	Low	Most Likely	High
CEE / blended (lean team)	€110,000	€168,000	€255,000
Western EU agency	€250,000	€370,000	€540,000
Mixed CEE + WEU lead	€155,000	€225,000	€330,000

Market Value Summary

Scenario	Estimated Range	Central Estimate
Pre-revenue IP / asset sale	€140K – €320K	€220K
Early traction (10–50K MAU, ad-free)	€350K – €900K	€550K

Scenario	Estimated Range	Central Estimate
Subscription PMF (€100–300K ARR)	€800K – €2.4M	€1.5M
Strategic acquisition (acqui-hire / portfolio)	€250K – €700K	€420K

⚠ **Biggest Uncertainty Driver:** Jupiter has **zero users and zero revenue** — it is a pre-monetisation MVP. With no installs, retention, or ARR signal, market value is anchored almost entirely to **replacement/IP value (€140K–€320K)**. A single proof point — a few thousand retained installs or even €10–30K ARR from a privacy-focused subscription — would shift the central estimate from the IP floor (~€220K) toward the €500K–€900K early-traction band.

2. Product Reconstruction

Reconstructed from direct inspection of the *com.jupiter.filemanager* source tree, *app/build.gradle.kts*, and the Gradle version catalog.

Technical Architecture

Layer	Technology	Version / Notes
Language	Kotlin	2.0
UI	Jetpack Compose + Material 3	Compose BOM
Architecture	MVVM + Clean Architecture	domain / data / feature
DI	Hilt (KSP) + Hilt-Work	—
Async	Coroutines + Flow	—
Navigation	Navigation-Compose	single-Activity
Persistence	DataStore Preferences + DocumentFile	SAF-aware
Security / Vault	Jetpack Security EncryptedFile + Biometric	AES-GCM, MasterKey
Background	WorkManager	automation rules
Build	Gradle 8.14.3 / AGP 8.7.3	minSdk 26 / target+compileSdk 35

Feature Module Map

Module	Key Capabilities
Browser & File Ops	Local browsing, copy/move/delete/rename, sort/filter, selection, SAF storage volumes
Remote / LAN Backends	SMB2/3 (SMBJ), SFTP (JSch), FTP/FTPS (commons-net), WebDAV (OkHttp) with a credential store
Wi-Fi Desktop Transfer	Embedded NanoHTTPD server for browser-based file transfer over LAN
Encrypted Vault	EncryptedFile AES-GCM entries + plaintext meta sidecar, biometric/Master-key unlock
Archive	zip/tar/gz/bzip2/7z (commons-compress + xz) and RAR extraction (junrar)
Cloud / Sync / Automation	Sync + automation repositories driven by WorkManager scheduling
Preview & Editor	In-app PdfRenderer viewer, image/video thumbnails (Coil), text editor
AI Assistant	Anthropic Claude Messages API, gated on a user-supplied key; NoOp fallback when absent
Cleanup & Analytics	Duplicate/merge detection, storage analytics, recent/downloads/favorites
Tags / Workspaces / Versions	Tagging, multi-workspace, file version history, activity log

3. Scope Decomposition

Complexity Ratings by Area

Feature Area	Complexity	Hidden Cost Drivers
Local File Ops & SAF	Medium-High	Scoped storage, SAF/DocumentFile, Android 11+ permission model
Remote / LAN Backends	Very High	Four protocols (SMB/SFTP/FTP/WebDAV), connection lifecycle, credential security, flaky-network handling
Wi-Fi Transfer Server	High	Embedded HTTP server, lifecycle, multipart upload, LAN discovery
Encrypted Vault	High	AES-GCM via EncryptedFile, MasterKey, biometric prompt, sidecar metadata
Archive Engine	Medium-High	Five formats, streaming extraction, large-file memory pressure
Automation / Sync	Medium-High	WorkManager constraints, idempotency, Hilt-Work integration
AI Assistant	Medium	Anthropic Messages API, key gating, graceful no-key fallback
Compose UI / Navigation	Medium-High	Material 3, 27 feature screens, state hoisting, theming

Complexity Multipliers

Factor	Impact	Rationale
Four real LAN protocols	+20%	Each backend (SMB/SFTP/FTP/WebDAV) is an integration project; testing requires live servers
Security-critical vault	+15%	Crypto correctness, key management, biometric flows carry zero error tolerance
AI-accelerated build	−30%	16 commits in one rapid push reflect heavy AI tooling; traditional teams take 2–3× longer
Thin automated test coverage	+15%	Unit tests exist for models/utils; backends and UI lean on manual QA
Android fragmentation	+10%	minSdk 26 → SDK 35: storage/permission behaviour differs widely across versions/OEMs

4. Methodology

Bottom-Up Estimation

The 33,932 lines of Kotlin across 212 files and ~27 feature packages were decomposed into 12 functional areas. Each area was estimated from code volume, number of integrated protocols/SDKs, UI interaction surface, and security/edge-case burden evidenced in the source.

Three-Point PERT

Optimistic / Most Likely / Pessimistic hours per area, combined as $E = (O + 4 \cdot M + P) / 6$, with $SD = (P - O) / 6$. Ranges are reported throughout — never single points.

Analogous Estimation

Calibrated against comparable Android file managers (Solid Explorer, X-plore, Cx File Explorer, Material Files, Files by Google) and public engineering norms for multi-protocol Android utilities.

Important Distinctions

- **Effort ≠ Duration:** ~3,159 person-hours can ship in ~7 months (2–3 FTE) or stretch to 18+ months (1 FTE).
- **Build Cost ≠ Market Value:** Replacement cost is the floor; installs, retention and revenue multiply it.
- **Code ≠ Product:** A shipped app also needs Play Store assets, store optimisation, support and onboarding not visible in the repo.
- **AI-assisted ≠ Traditional:** The 16-commit single-push build reflects AI-accelerated development; a conventional team would take 2–3× longer.

5. Team Composition

Required Roles

Role	Why Needed	Phase	Effort %
Senior Android Engineer	Architecture, Compose, Hilt, storage/SAF, remote backends	All	40%
Mid Android Engineer	Feature screens, archive, preview, automation, vault UI	2–8	27%
Backend/Network Engineer	SMB/SFTP/FTP/WebDAV integration, Wi-Fi transfer server	2–6	13%
UX/UI Designer	Material 3 system, iconography, onboarding, empty states	1–2, 4–5	9%
QA Engineer	Device-matrix testing, protocol/vault verification	3–8	6%
Product / PM	Scope, prioritisation, store readiness	All	5%

Delivery Team Options

Solo / Lean (1–2 people, 9–12 months)
1× Senior Android · 0.5× Designer
Lowest cost; matches a bootstrapped indie founder. Likely how this MVP was built.

Balanced (3 people, 6–8 months)
1× Senior Android · 1× Mid Android · 1× Backend/Network (shared Designer/QA)
Recommended for a funded indie or small studio. Parallel local/remote tracks.

Studio Delivery (5 people, 4–6 months)
2× Senior · 1× Mid · 1× Backend/Network · 1× Designer (shared QA/PM)
Maximum velocity. Suitable for agency delivery with hard deadlines.

6. Effort Estimate

Area-by-Area Breakdown (PERT)

Feature Area	Opt. (h)	ML (h)	Pess. (h)	PERT (h)
Local File Ops & SAF Storage	120	180	270	183
Browser UI & Core File Operations	220	320	470	318
Remote / LAN Backends (SMB/SFTP/FTP/WebDAV)	240	350	520	360
Wi-Fi Transfer Server (NanoHTTPD)	90	140	210	140
Encrypted Vault + Biometrics	110	160	240	161
Archive Engine (zip/tar/gz/7z/rar)	90	130	200	131
Cloud / Sync / Automation (WorkManager)	150	220	330	223
AI Assistant (Anthropic Claude)	70	110	170	113
Preview / Viewers / Editor (PdfRenderer)	120	180	270	183
Cleanup / Analytics / Tags / Workspaces / Versions	160	240	360	243
Compose UI / Theme / Navigation / Onboarding	180	270	410	273
DI / Architecture / Core / Result	70	100	150	102
Subtotal (coding)	1,620	2,400	3,600	2,430
+ 30% overhead (QA, PM, design, DevOps, docs)	486	720	1,080	729
TOTAL	2,106	3,120	4,680	3,159

Summary in Multiple Units

Metric	Low	Most Likely	High
Person-hours	2,106	3,159	4,940
Person-days (8h)	263	395	618
Person-months (160h)	13.2	19.7	30.9
Calendar months (2–3 person core)	5	7	11

7. Cost Estimate

Detailed Cost Model — Most Likely (CEE Blended Rates)

Role	Share	Hours	Rate	Cost
Senior Android Engineer	40%	1,264	€70/h	€88,480
Mid Android Engineer	27%	853	€48/h	€40,944
Backend / Network Engineer	13%	411	€58/h	€23,838
UX/UI Designer	9%	284	€45/h	€12,780
QA Engineer	6%	190	€40/h	€7,600
Product / PM	5%	158	€55/h	€8,690
Direct Labour		3,159		€182,332
Overhead (15%) − AI-build credit				−€18,000
Contingency (12%)				€19,720
TOTAL				€168,000

Cost Ranges by Scenario

Scenario	Low	Most Likely	High
CEE / blended (lean team)	€110,000	€168,000	€255,000
Western EU agency	€250,000	€370,000	€540,000
Mixed CEE + WEU lead	€155,000	€225,000	€330,000

8. Market Comparison

Comparable Android File Managers

Product	Model	Notes
Solid Explorer	Paid (~€2–3 one-off)	Closest power-user peer; cloud + LAN (SMB/FTP/WebDAV), strong UX. The bar Jupiter targets on power.
X-plore File Manager	Free + paid Donate	Dual-pane, deep LAN/cloud, Wi-Fi PC sharing; very feature-dense, dated UI.
Cx File Explorer	Free (ad-supported)	Clean UX, network (SMB/FTP/WebDAV) support; large install base.
Material Files	Free / open-source	Modern Material UI, archive + FTP/SFTP/SMB; closest design/architecture analogue.
Files by Google	Free (Google)	Massive reach, dead-simple UX, cleanup nudges; no real LAN/remote depth. The bar Jupiter targets on simplicity.
FX File Explorer	Free + Plus (~€2–4)	Privacy-positioned (no ads in Plus), network + media tools; closest privacy-stance peer.

Consumer Android Utility Value Drivers

Driver	Typical Signal	Relevance to Jupiter
Install base / MAU	100K–10M+ for top utilities	Currently zero — the dominant gap
Monetisation	One-off €2–4 or ad-supported	Not yet chosen; ad-free stance favours paid/subscription
Retention (D30)	15–30% for sticky utilities	Unknown — no telemetry
Differentiation	LAN depth, privacy, AI	Real LAN + vault + on-device-keyed AI = credible wedge

9. Market Value Estimate

Five valuation lenses are applied and triangulated. Because Jupiter is pre-revenue with no users, the estimate is anchored to replacement/IP value and converges on the project baseline of **€140K–€320K**.

Lens 1: Replacement / IP Value

Build cost at CEE blended rates is €110K–€255K (most likely €168K). As a clean, marketed IP asset with real LAN backends and an encrypted vault, a 1.0–1.3× multiplier applies. Result: **€140,000 – €320,000** — the floor, with zero revenue.

Lens 2: Comparable-Based Valuation

Comparable Android file managers monetise via one-off €2–4 purchases or ads. Pre-traction, comparables imply little above replacement cost; value only compounds with installs. At 10–50K engaged users an ad-free app could support **€350K–€900K**; at €100–300K ARR, **€800K–€2.4M**.

Lens 3: Feature-Depth Premium

Four real LAN protocols, an embedded Wi-Fi transfer server, an AES-GCM vault, five archive formats, and an optional Claude assistant exceed what most free file managers ship. Against the replacement floor this supports a **1.1–1.3×** depth premium — already folded into Lens 1's upper bound.

Lens 4: Risk Adjustment

Haircuts: zero users/retention proof (−15%), thin automated test coverage on backends/UI (−10%), single-contributor key-person risk (−10%), unproven monetisation (−10%). Offsetting: modern Kotlin/Compose/Clean stack (+10%), privacy/no-ads positioning (+5%), CI building debug+release APKs (+5%). Net: ≈ −25%, which keeps the central estimate near the IP floor.

Lens 5: Strategic / Acqui-Hire Value

To a privacy-focused app portfolio, an OEM, or a NAS vendor (e.g. Synology/QNAP-adjacent), the LAN stack + vault + clean architecture have acqui-hire value of **1.5–2.2× IP** for the team and codebase. Range: **€250K – €700K** in a strategic deal.

Final Market Value Ranges

Scenario	Range	Central Estimate
Pre-revenue IP / asset sale (current state)	€140K – €320K	€220K
Early traction (10–50K MAU, ad-free)	€350K – €900K	€550K
Subscription PMF (€100–300K ARR)	€800K – €2.4M	€1.5M
Strategic / acqui-hire deal	€250K – €700K	€420K

Scenario	Range	Central Estimate
Triangulated current value	€140K – €320K	€220K

Triangulated Current Value: €140K – €320K (central ≈ €220K). As a pre-revenue, zero-user MVP, Jupiter is valued essentially at replacement/IP cost. Demonstrating retained installs or modest ARR is the single highest-leverage way to move into the €350K–€900K early-traction band.

10. Assumptions and Limitations

Hard Evidence (Known)

✓ Full source: 212 Kotlin files, 33,932 lines of Kotlin, 237 files total (repo scan)

✓ Stack confirmed: Kotlin 2.0 / Compose / Material 3 / MVVM + Clean / Hilt / Coroutines; Gradle 8.14.3 + AGP 8.7.3; minSdk 26 / target+compileSdk 35

✓ Real LAN backends in code: SMBJ (SMB), JSch (SFTP), commons-net (FTP), OkHttp (WebDAV); NanoHTTPD Wi-Fi server

✓ EncryptedFile/MasterKey vault, WorkManager automation, PdfRenderer viewer, archive (commons-compress + xz + junrar)

✓ Anthropic Claude assistant gated on a user key with a NoOp fallback; JUnit unit tests; CI builds debug + release APKs

✓ Git: 16 commits, single rapid push; version 0.1.0

Inferred (Not Directly Observed)

~ Build effort (estimated from code volume, protocol count, and security complexity)

~ Rate assumptions (CEE blended €45–€75/h market norms)

~ Comparable products and their monetisation (public store listings)

~ AI-acceleration credit on cost (inferred from 16 commits in one push)

Unknown / Missing

✗ Users / installs / MAU — none; the app is unreleased

✗ Revenue / ARR / monetisation model — not yet chosen; most dominant valuation variable

✗ Retention, crash-free rate, store ratings — no telemetry or Play Console data

✗ Real signing keystore — release currently signed with the debug key (placeholder)

11. Recommended Next Steps

For Valuation Uplift (Sale / Fundraising)

→ Ship to Play Store and instrument installs/retention — even 5–20K engaged users changes the valuation band

→ Pick a monetisation model (one-off unlock or privacy-first subscription) and prove €10–30K early ARR

→ Publish a landing page and ASO assets emphasising the no-ads / privacy / real-LAN wedge

For Technical Hardening

→ Replace the debug signing config with a real release keystore before publishing

→ Expand automated coverage to the four remote backends and the vault (highest-risk, lowest-tested areas)

→ Add crash reporting and privacy-respecting analytics to generate the retention evidence valuation needs

For Product Differentiation

→ Lean into the Claude assistant as a marketed feature (smart rename, cleanup, search) with clear key-bring-your-own UX

→ Polish onboarding to hit the 'as simple as Files by Google' bar for non-power users

→ Document NAS/LAN setup flows to win the LAN/NAS persona against Solid Explorer / X-plore

12. Appendix

A. Domain Repository Inventory

Repository	Responsibility
FileRepository	Local file listing and operations
RemoteAccessRepository	SMB/SFTP/FTP/WebDAV remote access
ConnectionRepository	Saved remote connections + credentials
VaultRepository	Encrypted vault entries (EncryptedFile)
TransferRepository	Wi-Fi / desktop transfer sessions
AutomationRepository	WorkManager-driven automation rules
SyncRepository	Folder sync
StorageAnalyticsRepository	Storage usage / cleanup analytics
TagRepository / BookmarkRepository	Tagging and bookmarks
WorkspaceRepository / VersionRepository / ActivityRepository	Workspaces, file versions, activity log

B. Confidence Assessment

Dimension	Confidence	Rationale
Build effort estimate	High (±20%)	Full codebase; clear architecture and protocol surface
Cost estimate (CEE)	Medium-High (±25%)	Blended rate data available; AI-build credit inferred
Market value (pre-revenue)	Medium (±40%)	No users/revenue; anchored to IP floor
Market value (with traction)	Low-Medium (±50%)	Monetisation model unchosen; comparables are one-off/ad-based

This report was produced by AI-assisted technical due-diligence combining direct repository inspection of the Jupiter Android codebase and external market research on consumer Android file managers. It is for informational purposes and does not constitute financial advice.